The De Broglie Relation for the Magnetron RF Field

1. The photon limit

For a photon the rest mass vanishes, $m_0 = 0$, so the relativistic energy–momentum relation

$$E^2 = (pc)^2 + (m_0 c^2)^2$$

reduces to

$$E = pc \quad \Rightarrow \quad p = \frac{E}{c}.$$

Combining this with Planck's law, $E = h\nu = \frac{hc}{\lambda}$, gives the momentum of a light quantum,

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda},$$

which is the **De Broglie relation for a massless particle**,

$$\boxed{\lambda = \frac{h}{p}}.$$

2. Extension to particles with mass

De Broglie postulated that the same wave–momentum correspondence holds for a particle carrying rest mass $m_0 \neq 0$:

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

For an electron, applying Planck's law $E = h\nu$ yields only an **approximate** dispersion, since the momentum of a massive particle is not simply E/c; the equality $p = E/c$ becomes exact only in the ultrarelativistic ($v \to c$) limit.

3. The magnetron RF quantum

We now associate a De Broglie wavelength with the magnetron radio-frequency (RF) field. In this framework the quantum of action carried by the magnetron mode is a distinct constant, denoted $h_{m,\mathrm{RF}}$, playing the role of an effective Planck constant for the RF sector:

$$h_{m,\mathrm{RF}} \simeq 5.663809031 \times 10^{-9}\ \text{eV·s}.$$

A particle (or field mode) propagating at speed c then has an associated wavelength

$$\lambda = \frac{h_{m,\mathrm{RF}}}{p}.$$

3.1 Nonlinear energy law

The full frequency dependence of the RF energy is described by the fitted relation

$$E(\nu) = -0.0729133537 \ + \ 373.701413935296492 \ln(1 + 1.580487370666666666653 \times 10^{-11}\,\nu) \\ - \ 1.549823289304468027271 \times 10^{-19}\,\nu^2,$$

with E in eV and ν in Hz. The logarithmic term governs the low-frequency (linear) regime, while the quadratic term introduces a high-frequency correction.

3.2 Linear approximation

In the linear (radio-frequency) region the energy law reduces to the Planck-type form

$$E_m \simeq h_{m1}\,\nu,$$

so that $h_{m,\mathrm{RF}}$ is treated as constant and the De Broglie relation retains its simple form,

$$\boxed{\lambda = \frac{h_{m,\mathrm{RF}}}{p}}.$$

4. Klein–Gordon starting point and the massive dispersion

For particles with rest mass $m_0 \neq 0$ we begin from the Klein–Gordon equation

$$\left(\Box + \frac{m_0^2 c^2}{\hbar^2}\right)\psi = 0, \qquad \Box = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2,$$

whose plane-wave solutions reproduce $E^2 = (pc)^2 + (m_0 c^2)^2$. In the linear RF approximation Planck's law takes the form

$$E = h_{m,\mathrm{RF}}\,\nu = h_{m,\mathrm{RF}}\,\frac{c}{\lambda},$$

and the De Broglie wavelength follows as $\lambda = h_{m,\mathrm{RF}}\,c/E$.

5. Numerical check

Evaluating the linear relation $E = h_{m,\mathrm{RF}}\,\nu$ in the RF region:

$$E = (5.6638 \times 10^{-9}\ \text{eV·s})\,(5.754173859 \times 10^{8}\ \text{Hz}) \ \approx \ 3.259\ \text{eV}.$$

This confirms the internal consistency of the linear approximation:

$$3.259\ \text{eV} \ = \ 5.6638 \times 10^{-9}\ \text{eV·s} \ \times \ 5.754173859 \times 10^{8}\ \text{Hz}.$$

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